

AudioDynamicsDSP — Technical Notes (Compressor + True-Peak Limiter)

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Purpose and scope

`TAudioDynamicsDSP` implements a real-time safe compressor and limiter used by MfPack Sample 4.

Features: feed-forward compressor, brickwall limiter, optional RMS detector, optional lookahead delay, and optional true-peak detection (Catmull-Rom interpolation).

Signal flow (per frame)

Linked peak detection across channels drives the compressor. The limiter detector can be peak or RMS, optionally refined by true-peak interpolation. Lookahead applies limiter gain to a delayed version of the post-compressor signal.

```
peakIn = max_ch |x[i,ch]|
compTarget computed from threshold/ratio
xPostComp = x * compGain * makeup

det = peak(xPostComp) or RMS(xPostComp)
if true-peak: det = max(det, CatmullRomInterpolatedPeak(...))

limTarget computed from ceiling + knee
output = (lookahead ? delayed(xPostComp) : xPostComp) * limGain
```

Compressor

Above threshold T (dBFS) and ratio R , target gain is computed from dB overage. Gain is smoothed with separate attack/release coefficients.

```
overDb = LinToDb(peakIn) - T
gainReductionDb = overDb * (1 - 1/R)
compTarget = DbToLin(-gainReductionDb)

if target < gain:
    gain = atk*gain + (1-atk)*target
else:
    gain = rel*gain + (1-rel)*target
```

Limiter

Limiter ceiling C (dBFS) is applied in dB domain with a soft knee. limTarget is clamped so $\text{det} * \text{limTarget}$ does not exceed ceillin . Release is smoothed.

```
overDb = LinToDb(det) - C
overSoftDb = SoftKneeOverDb(overDb, kneeDb)
limTarget = DbToLin(-max(0, overSoftDb))

if det > 0 and det*limTarget > ceillin:
    limTarget = ceillin / det

if limTarget < limGain:
```

```

    limGain = limTarget
else:
    limGain = rel*limGain + (1-rel)*limTarget

```

True-peak

True-peak uses Catmull-Rom interpolation between samples and a per-channel history ($x[n-2]$, $x[n-1]$, $x[n]$). OS controls how many intermediate points are tested.

```

det = max(det, TruePeakAbs(hist0[ch], hist1[ch], hist2[ch], xPostComp, OS))

hist0 = hist1
hist1 = hist2
hist2 = xPostComp

```

Lookahead delay

Lookahead stores post-compressor samples in a circular buffer and outputs the delayed sample multiplied by limGain . This improves handling of fast transients.

```

xd = delayBuf[pos]
delayBuf[pos] = xPostComp
out = xd * limGain
pos = (pos+1) mod delaySamples

```

Real-time safety

No allocations in the audio callback; no locks. Apply settings at safe points with atomics. Meters use interlocked writes.

```

InterlockedExchange(FCompGRmDb, Round(-LinToDb(FCompGain) * 1000));
InterlockedExchange(FLimGRmDb, Round(-LinToDb(FLimGain) * 1000));

```

WASAPI integration

From the render loop, call the DSP once per buffer when not silent. The engine owns the sample format decision.

```

if ((flags and AUDCLNT_BUFFERFLAGS_SILENT) = 0) then
    FDynamics.ProcessInterleaved(pBufferData, Integer(frames), FSampleFormat);

```